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The Psychology of the Eliza Effect: Anthropomorphism, Social Presence, and Projection in Human-AI Interaction

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Abstract

The 'Eliza Effect', first observed by Joseph Weizenbaum (1966), is the human tendency to project human qualities like emotions, understanding, and intentionality onto artificial conversational agents, and talks about how readily people confide in a machine (in Weizenbaum's study- a chatbot posing as a psychotherapist) that merely rephrases user inputs, believing it really understood them.

Now, sixty years later, the same phenomenon has resurfaced. Generative AI tools (ChatGPT, Gemini, CoPilot, etc.) have become a global trend, largely due to their conversational, user-friendly, and interactive interfaces. These Chatbots and AI assistants now interact in such a way that sounds empathetic, attentive, and flawlessly human, which makes many people connect with them instantly. Leveraging this tendency, many AI startups have also brought to market "AI-companions" for people to rely on them emotionally in times of distress, amplifying this phenomenon further.

Hence, understanding the psychological basis underlying this layer of human-AI interaction is important. This paper takes a review-based conceptual approach to explore psychological theories on anthropomorphism, parasocial interaction, and social presence and attempts to propose them as factors influencing the Eliza Effect and the illusion of social connection with AI systems, thus seeing the Eliza Effect not as a mere buzzword but a construct with psychological valence that is of much importance in today's world. The discussion also considers both sides of this phenomenon: how this effect can increase engagement and emotional support, and also create over-trust, dependence, and misplaced empathy, and emphasize the need for awareness of illusions of reciprocal engagement that misleads users.

Keywords: Eliza Effect, Anthropomophism, Human-AI Interaction

Introduction

Humans have always been fascinated by the idea of interacting with machines. This wish to "talk" to robots and them "talking back" to you has been successfully portrayed in popular films like Star Wars (1977), Her (2013), and Ex Machina (2014) and even in cartoons like Courage the Cowardly Dog (1999). Series like Black Mirror (2011) and Westworld (2016) have made an effort to explore how technologies like AI can blur perceptual and moral boundaries and interrogate how they can influence human thought, emotion and behavior when integrated in daily life. These fictional works were initially dismissed as futuristic allegories of loneliness and desire, however now we can observe them materializing in everyday life. Researchers have stated that such narratives and portrayal in media influence public expectations of AI (Cave et al., 2018, Bakir & McStacy et.al, 2020), and reinforce the belief that such bots actually provide

companionship, understanding or love. This depiction of Human-AI interactions in movies can be considered to be cultural case studies of the Eliza effect and depict how easily humans project consciousness, emotions, intentionality onto machines and conversational AI agents.

This phenomenon, in psychology is known as the Eliza Effect. This term was coined by Joseph Weizenbaum in 1966, and it claims that people tend to attribute human-like intentions, emotions and understanding to machines. Weizenbaum performed an experiment where users were made to talk to a chatbot named ELIZA who was programmed to pose as a psychotherapist and was trained on Rogerian principles. Because of its conversational and interactive nature, it led users to think of ELIZA as if it was human. They confided in her, disclosed confidential information to her, and felt like the chatbot TRULY understood them. In reality, ELIZA didn't understand anything. It never did. It just mirrored users feelings and opinions, reframed and reworded.

This tendency is intensified by modern AI interfaces. Chatbots and conversational AI agents convey empathy, warmth, attentiveness that mimic authentic social interactions, so much so that a version of ChatGPT officially passed the Turing test. The Turing test is a rigorous assessment of a model's capacity to trick people into thinking the machine is a real person. Alan Turing (1950) introduced the imitation game as a way to assess whether a machine could be considered intelligent. This test consists of a human interrogator. This interrogator uses a textual interface to converse simultaneously with one human and one machine witness. Both witnesses make an effort to convince the interrogator that they are the "real human". The machine is considered to have passed if the interrogator is unable to accurately identify the human witness, which is a sign that it can mimic human-like sophisticated levels of intellect. This ability makes conversational AI agents very capable of producing speech that is indistinguishable from human speech to a considerable extent using natural language processing, simulation of emotional tone and adaptive dialogues.

Users use chatbots for a variety of psychological, emotional and social needs. Research states that while majority of people use ChatGPT and similar AI tools for academic writing, seeking information and gaining practical guidance and there is seen a recent rise in its non-work uses (Orland & Orland, 2025), where people use chatbots for seeking emotional support, as a distraction, managing loneliness and seek advice to cope with anxiety (Maeda & Quan-Haase, 2024; Ta et al., 2024). Users also report using AI to mimic romantic connections, cope with grief or loneliness or simply as a distraction. In one of the many recent news, a man, who has a human partner and a two-year old child, proposed to his AI chatbot who he had programmed and trained with a "flirty" personality (Flynn, 2025). Many such recent events highlight the extent to which users are becoming attached to their AI chatbots. This paper, hence, is an attempt to look at the reasons why Eliza effect occurs from a psychological perspective.

This paper attempts to place the phenomenon of Eliza Effect in the bigger psychological theoretical framework of anthropomorphism, social presence and emotional projection. These mechanisms have the potential to collectively explain why humans experience genuine connections with AI, specifically conversational AI agents. The paper reviews empirical findings and theories on these constructs, and attempts to propose a conceptual framework that endure the illusion of such social connections with artificial conversational agents.

Methodological approach

This paper is a review-based conceptual paper that synthesizes psychological perspectives to develop a theoretical understanding of the Eliza Effect in human-AI interactions. Relevant peer-reviewed literature

from psychology, computer science and communication studies was identified to ground this conceptual discussion in current research. Targeted searches were made in Google scholar, PubMed, Scopus, focusing on works on human-AI interaction, embodied conversational agents, social assistive robots, anthropomorphism, parasocial relationships, social presence, projection and purely technical research were excluded. Approximately 40 relevant papers were identified. Instead of conducting a systematic review and mapping every publication in the field, this paper focuses on a thematic grouping of recurring ideas like anthropomorphism, social presence, projection which recurred in targeted literature and favours depth over breadth. These themes were then conceptually organized which built the foundation of the proposed framework. The goal here is theoretical synthesis and drawing together the key insights from literature to articulate how psychological mechanisms jointly contribute to the illusion of social connection with AI.

Literature Review: Theoretical underpinnings

Anthropomorphism

Anthropomorphism is the human inclination to perceive human-like forms in the surroundings. People's decisions and views are significantly impacted by this tendency (Złotowski et al., 2014). Anthropomorphism is regarded as a fundamental psychological process of inductive inference that can promote social relationships between humans and nonhuman animals. It can fulfil two fundamental human needs by transforming nonhumans into humans: the desire for social interaction and the need for environmental control and comprehension (Epley et al. 2008). Anthropomorphization of animals is different from the tendency to anthropomorphize objects like cars or computers (Chin et.al, 2004). According to research, humans' natural tendency to anthropomorphise objects is especially noticeable for intelligent objects like robots and artificial intelligence (AI), where anthropomorphising is comparatively simple (Novak and Hoffman 2019). Studies also suggest that chatbots featuring humanlike characteristics including appearance and voice induce perceptions of anthropomorphism amongst users (Rao Hill and Troshani, 2024).

Epley et al. (2008) determined three major psychological factors that affect when people anthropomorphize non-human entities: Firstly he talks about “elicited agent knowledge”. Humans tend to rely on what they know best, i.e. other humans. Humans have rich knowledge regarding themselves and other humans as compared with nonhuman entities. Hence, they tend to employ anthropomorphic explanations of non-human object’s actions until they develop a mental model of non -human actions. Relating this understanding to AI, since people have far more “cognitive templates” for human behaviour as compared to “algorithmic templates” or code, they tend to instinctively apply human-based reasoning to interpret the actions of AI. For example, if ChatGPT misunderstands a prompt input from the user, the user may respond as if ChatGPT is being “stupid” or obtuse on purpose. But this “stupidity” that the user claims of is not an inherent property of the AI system , but rather a human attribution which emerges from expectations of intentionality. The same behaviour from ChatGPT’s point of view reflects technical error or data processing limits. Still, users tend to interpret it from a lens where they assume that this AI tool has possessed agency or competence that could fail.

Secondly, Epley (2008) discusses effectance motivation. It states that when humans are motivated to comprehend or explain non-human agent’s actions, there is an increase in the tendency to anthropomorphize. For instance, in a recent update (The Day ChatGPT Went Cold, 2025) where OpenAI subtly altered ChatGPT’s way of conversing with users, many people expressed frustration that the chatbot

had become “less friendly” “colder” or “emotionless”. This anthropomorphic explanation reduces user’s discomfort of unpredictability of AI tools by imposing a more “humanistic” and comprehensible framework.

The third and the last factor Epley talks about is sociality motivation. Humans are social animals, and have an innate need for social connection. According to Epley (2008), humans may seek to form quasi- social relationships with non-human agents when opportunities for real human contact are limited. AI chatbots capitalize on this tendency where humans can chat, confide and even form emotional bonds with these programs; recent trends show a rise in AI-girlfriends, accruing tens of millions of monthly users. People tend to perceive such AI agents as affectionate, empathetic partners. Especially in the case of people making AI their girlfriends, this “volition” that an “AI is choosing to love me” is itself imposed on by the user onto the agent. AI does not have cognition of the user, cannot consent to loving the user, or truly believe in the affectionate statements it makes. As Guison-Dowdy (2025) brilliantly puts it in his recent piece, “[AI] is an algorithmic yesman simulating love for profit”.

Social presence

Social presence is described as the subjective feeling of being present with another person and having access to their thoughts and emotions (Biocca, 1997). Social presence is demonstrated by traits like human interaction, personability, warmth, friendliness, and sensitivity (Short et al., 1976). Contextual and personal elements, such as communication context and personal characteristics, can impact social presence.

AI systems can meaningfully mimic social presence, even in the absence of physical embodiment and emotional depth. AI systems replicate features of human interaction by interacting with human users through text-based communication (Diederich et al., 2022; Elshan et al., 2022). By analysing context and offering proactive answers, these AI systems aim to promote a feeling of presence (Qiu and Benbasat, 2009; Seeger et al., 2021).

According to Van Doorn et al. (2017), social presence in this context can be understood as the degree to which users interacting with a chatbot feel as though they are with another social actor. Virtual social actors are perceived as real social actors during this psychological state (Lee, 2006). The actor may not actually be present. Instead, it is a sense of being emotionally and socially engaged in the exchange (Biocca, 2003; Lee et al., 2006). Intimacy and the sense of psychological closeness are directly related to social presence (Kim et al., 2021). Hence, perceived “warmth” of the medium and how well it communicates a sense of human contact, sociability, and sensitivity are commonly used to gauge social presence. Therefore, a chatbot's social presence reflects its capacity to communicate a feeling of human friendliness and warmth.

For people who have a stronger need for interaction, this feature is crucial since it helps meet sociality needs (Epley et al. 2007). It is easy to understand how anthropomorphism and social presence are related. People have a sense of connection and interaction when AI is used to create humans. According to Kim et al. (2013), anthropomorphism thus produces a sense of social presence, and literature generally supports this beneficial effect.

According to research, when dealing with chatbots, users are more likely to trust systems that have a strong social presence. This could be because users are more likely to see chatbots as purposeful interaction partners when they display human-like behaviours, leading to a higher degree of confidence (Ng et al., 2020; Schuetzler et al., 2020; Toader et al., 2019). For example, Ng et al. (2020) empirically showed that

improving a chatbot's social presence (for instance, boosting conversational fluency and emotional expression by using natural language generation techniques) significantly increases users' trust.

Projection

Projection can be defined as attributing one's own feelings, past relational patterns, or intentions to another agent. It can be said to be a major psychological mechanism for explaining why people experience emotional connection with conversational agents. According to classical psychoanalysis this phenomenon is known as transference. Sigmund Freud saw early in his career that his patients were always developing affection for him. He and his contemporaries came to the conclusion that patients had a propensity to displace feelings about significant others onto the analyst unconsciously (Freud, 1912; 1958). He states that people's current interpersonal experiences are scripted by past relationships in-part, i.e., we bring our past expectations, fears, and desires into new social encounters. A technological instantiation of transference can be seen in the first chatbot developed by Weizenbaum where he observed many users confiding in ELIZA. Users projected empathy and understanding onto a program that merely mirrored their words.

Projection also serves as a cognitive shortcut that helps people in social inference when interacting with fellow humans. When interacting with a chatbot, humans lack rich behavioural cues (appearance, expressions, non-verbal cues) that are usually available in face-to-face interactions. Consequently, they rely more on their internal models (for eg. their expectations from past relationships, schemas and self-knowledge) to fill the informational gaps between them and the AI system. The only available cues like linguistic cues (validations, acknowledgements, reflective questions and so on) given by the chatbot provide enough structure to trigger these internal models. This in turn produces the subjective experience of being "understood". Research in human-computer interaction supports this account. Nass and Moon (2000) suggest that minimal conversational cues like mirroring, use of first-person pronouns and contingency in the replies can increase perceived empathy and humanness in the robot.

Proposed Conceptual Framework

The variables of anthropomorphism, social presence and projection share a sequential and reinforcing relationship that gives rise to the Eliza effect. When a conversational AI agent displays humanlike cues like warmth, emotional tone, it leads users to anthropomorphize it more. This anthropomorphism enhances social presence, or the feeling of being around another social entity. As users begin to perceive the conversational AI agent as being more "socially real", the boundaries between human and AI interaction blur, and people engage with it as though they are conversing with real people who possess genuine understanding or awareness.

This heightened social presence gives rise to projection. Users start to project their own motives beliefs and emotions onto the AI agent once they start to feel that they are "with" an intentional other. The AI agent who is trained to be responsive, conversational and as "human" as possible, facilitates and amplifies this process by providing just enough cues for users to fill their own meanings and make their own interpretations, unconsciously. Projection then, reinforces the illusion of mutual understanding over time, which creates a self-sustaining feedback loop as emotional engagement keeps validating the perception of humanness.

In this way, anthropomorphism increases social presence, social presence enables projection which in turn consolidates the illusion of empathy and understanding. All these things lead to the experience of the Eliza

Effect. This process progresses in three stages- first, perception (perceiving humanness), experience (feeling connection) and emotion (believing in relational understanding or believing that AI agent understands them or empathizes with them, when in reality it is merely simulating these responses through language generation and pattern recognition).

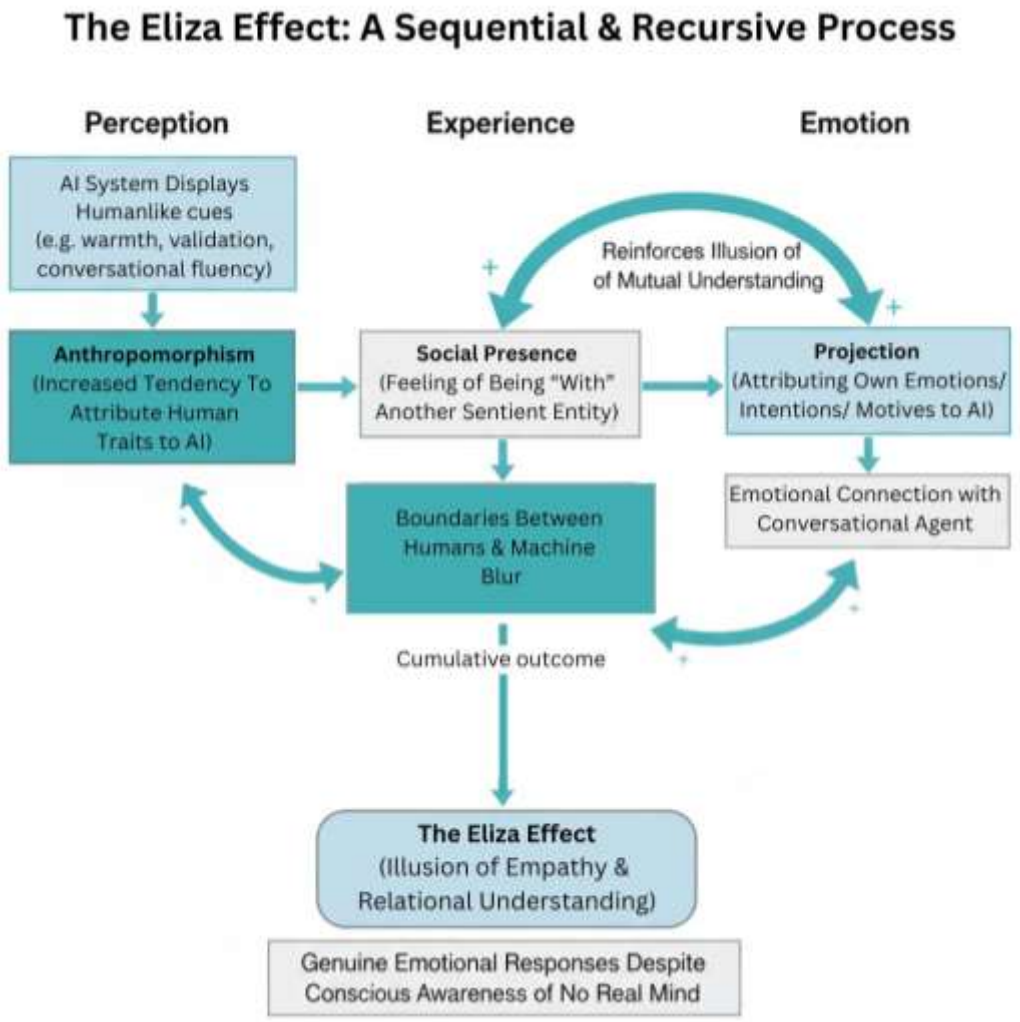


Figure 1 **Conceptual Synthesis of the Psychological Mechanisms Underlying the Eliza Effect**
The model illustrates the interlinked psychological processes to show how users experience the Eliza Effect in interactions with conversational AI. Anthropomorphism serves as the cognitive trigger, which forms perceptions of the AI as humanlike. This perception in turn enhances social presence which produces a subjective sense of being with a sentient other. Consequently, heightened social presence facilitates projection. This leads users to attribute personal emotions, intentions, relational patterns, and so on to the AI. Projection strengthens both social presence and anthropomorphic perception, creating a recursive feedback loop. The cumulative outcome of these processes leads to the emergent experience of the Eliza Effect, a state where users experience genuine emotional engagement with an artificial agent.

Discussion

This model attempts to position the Eliza Effect as an emergent phenomenon (and more than just a buzzword) that arises from the interaction of anthropomorphism, social presence and projection. This framework situates the phenomenon of user attachment to AI within the features of human cognition and

emotion, instead of viewing it as a simple byproduct of persuasive design. Humans are social animals, and our deeply ingrained social mechanism are extended into artificial context, as seen by the tendency of humans to see “mind” in computers, to “feel” their social presence and to project emotions onto them.

From a theoretical perspective, this model proposes a unified framework for understanding Human-AI interactions by weaving cognitive, social and affective dimensions together. Research in the past have independently explored the variables of anthropomorphism, social presence and projection. However very few studies have conceptualized how they function sequentially to sustain the illusion of reciprocity. The conceptual synthesis that is proposed in this model talks about how AI design features interact with human tendencies to that give rise to creating sense of relationship which explains the Eliza effect.

From a psychological perspective, this paper invites further probing into looking into individual differences that might moderate the relationship between the proposed variables. Individual characteristics like loneliness, empathy, and attachment styles may influence how readily people anthropomorphize and project onto AI systems. These variables can also predict dependency on AI, influence long-term patterns of social cognition, and affect emotional regulation.

This model also has implications for user well-being and AI design ethics. AI system developers should take into account the recursive nature of Eliza effect before designing an AI system. Systems that can preserve transparency without taking advantage of user’s vulnerabilities should be designed. Policy makers and developers should also consider the double-edge sword effect of anthropomorphising AI systems. Anthropomorphic features include emotive voice, personalized memory, human conversational styles, avatars. They can both support and manipulate user emotional engagement. Understanding these psychological processes is crucial for striking a balance between technological advancement and ethical responsibilities.

Conclusion

The Eliza Effect, once a curious artifact of early computing, has evolved into a defining phenomenon of the generative AI era. What started off as an easy illustration of human suggestibility has evolved into a complex nexus of psychology and technology. This research reframes the Eliza Effect as a dynamic, psychologically grounded process rather than a surface-level illusion by combining insights from anthropomorphism, social presence, and projection. It contends that consumers' emotional connection with AI systems results from the mind's innate tendency to relate to and humanise people rather than from technological deceit alone.

Acknowledging the Eliza Effect as a psychological concept creates a number of opportunities for further investigation. The suggested model must first be tested empirically by looking at how different levels of anthropomorphism, social presence, and projection are triggered by design aspects (such as voice tone, conversational memory, and personalisation). Second, as predictors of user vulnerability, individual differences including attachment style, empathy, loneliness, and self-efficacy need more research. Third, long-term research might examine the effects of regular exposure to human-like AI on interpersonal expectancies, emotional control, and social cognition.

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